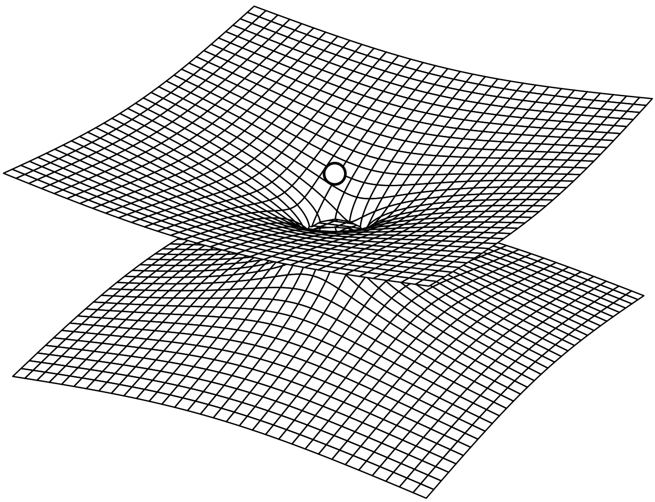

Lecture Notes on General Relativity

Physics 225

Vercil Juan

IV - BS Physics

1st Semester, AY 2026-2027

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\sigma\nu} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu})$$

$$\frac{d^2 x^{\sigma}}{d\lambda^2} + \Gamma^{\sigma}_{\mu\nu} \frac{dx^{\mu}}{d\lambda} \frac{dx^{\nu}}{d\lambda} = 0$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

"Rationem vero harum Gravitatis proprietatum ex Phenomenis nondum potui deducere, et Hypotheses non fingo."

– I. Newton

Preface

Relativity. When I was a child, I remembered wondering why things go down the way they do. My folks and those who are older just answer to me: That's simply the way things are. Later on, when I was older, I picked up this old book called *Science in Daily Life* where I learned much about science and chemistry and all those good shit. That book was one of my prized posessions, I still have it today, in fact. When I was a little older, right around fourth grade, my neighbor was throwing out books she no longer used, and part of that mixture were books on chemistry and books on Physics.

It was there where I learned about Newton and his apple. It was there where I learned about Einstein and binding energies and logic gates, of nuclear physics and atom bombs; where I learned that gravity is a force that drew upon it other masses by the product of their quantities the inverse of the square of their distances. That made me interested in orbital mechanics, astronomy, the Apollo program. That book was one of the reasons why I am where I am today. I say all this, because we all have an origin story of some sort. Some story we call ours, that tells our own humble beginnings. This was a story of how I fell in love in Physics, and why I chose it to be my career once I reached adulthood. That's why I decided to be a Physicist.

Newton had origin stories. Einstein had an origin story. We must treasure where we came from, and what it took for us to get there, is all I'm trying to say. Anyways, I first leard about general relativity all those years ago, but thought it was very difficult and a hassle to understand. Understanding special relativity and its implications were much easier, but boy, GR is a whole 'nother level. But fascinating! Differential geometry, manifolds, paradoxes, black holes, all so fascinating and interesting for me, part of the reason why I fell in love with the damn philosophy of reason. But also, scary, all those numbers and derivatives and the like, all so intimidating, that I thought back then I would never even learn it. And so, I find it funny, now, as an adult, that I took it as my undergraduate specialty of choice, joining a laboratory that specialized in gravity, taking my thesis on gravity. The amount of difficult mathematics one had to master to reach this point is insane, but, much like a mountain, the difficulty is part of the journey, and to reach the top is sweeter than the pleasures we get to enjoy in our daily life.

And so, here I am today, taking a graduate course in my undergraduate degree, about the subject that fascinated me as a child, but I found too difficult to climb and learn. Now, I know enough to say that what I know is a drop in an endless ocean, and relativity, the jewel of modern physics, in both beauty and elegance, is the pearl at the bottom of it. It makes me glad that it is what I chose to dedicate my academic life to, and I find within me the desire to add to the sum of human knowledge, to leave a legacy behind and be remembered in such way. It is only recently that I learned to appreciate physics for what it is, a beautiful description of reality. In my early college years, the stress of it all consumed me, and now I am about to graduate, I look back think of all those things I learned, and it impresses me that I could learn so much.

Anyways, these lecture notes are precisely that, transcriptions of lectures on the course of General Relativity, by one of the leading physicists in the country, Dr. Vega, whom I personally idolized since I learned about him when I was in highschool, being a homegrown gravitational physicist and all that, and now I'm studying under him, ain't that something. The content is precisely what he talks about in the lecture. Although not exactly verbatim, I tried my best to record all the words being said and tried to capture the topic being discussed at the moment, doing my best to convey the idea and meaning of the lecture. So that, in the future, I may look upon it when I forget something or simply as I please, and others, peers, students, friends, which I may give these lecture notes to, may understand it in a manner that I understand it today. I wish the future may be kind to us, and when we look back, to our stories and origins, may we smile and reminisce in mild happiness, and be proud of what we have become.

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Physica Vincit Omnia

Physics 225 - General Relativity 1
Course Information

Course Description

Description: Manifolds, modern differential geometry and tensor analysis; basic principles of general relativity; Einstein field equations and their mathematical properties; exact solutions; linearized theory; variational principles and conservation laws; equations of motion; gravitational waves; experimental tests.

Instructor Details

Instructor: Dr. Michael Francis Ian Vega

Email: ivega.upd.edu.ph

Lecture Hours: WF, 10:00 - 11:30 AM

Course Requirements

Problem Sets Problem sets will be assigned throughout the semester, ideally one per week. These will be submitted electronically using a Google Form that I will be sending out close to the deadline. From each problem set a random selection of items will be graded. It is your responsibility to ensure that your work is neat and legible. If we cannot understand your work then you will not get any credit. These problem sets will make up the 30% of your final grade.

Written Exam There will be two written examns in the course. Each exam will constitute 30% of the final grade. These are in-class, open-notes exams. You will have to designate one notebook as your 225 notebook at the start of the semester. Only this notebook may be used to help you during the exams.

Oral Exam This will be scheduled during or shortly after Finals Week. This will make up the remaining 10% of your final grade.

Main References

We shall not have a main textbook for the class. I used *Schutz*, supplemented by the first edition of *d'Inverno*, when I first started learning GR. I think they are both very good texts. However, I have my own way of presenting GR that will not always follow the structure or presentation of any single book. My approach is closest in style to the treatments of *Liang+Zhou* and *Wald*.

There are numerous sources you can refer to. I highly recommend that you look at different texts to help understand my lectures better. Here are some good books I've used on numerous occasions. - I. Vega

Course Syllabus

The Syllabus for the course can be found here:

- Course Syllabus

For General Relativity

- R d'Inverno, J Vickers, **Introducing Einstein's Relativity: A Deeper Understanding**, 2022
- C Liang and B Zhou, **Differential Geometry and General Relativity**, 2023
- R Wald, **General Relativity**, 1984
- V Ferrari, L Gualtieri, P Pani, General Relativity and its Applications, 2021
- T Padmanabhan, Gravitation: Foundations and Frontiers, 2010
- A Zee, Einstein Gravity in a Nutshell, 2013
- S Carroll, Spacetime and Geometry: An Introduction to General Relativity, 2003
- Choquet-Bruhat, Introduction to General Relativity, Black Holes, and Cosmology, 2015

- J Hartle, Gravity: An Introduction to Einstein's General Relativity, 2003. (undergrad)
- B Schutz, **A First Course in General Relativity 2nd Ed.**, 2009. (undergrad)
- MP Hobson, GP Efstathiou, and AN Lasenby, General Relativity: An Introduction for Physicists, 2006. (strong for cosmology)
- S Hollands, K Sanders, **Lecture Notes on General Relativity**, 2015
- S Weinberg, **Gravitation and Cosmology**, 1972

For Differential Geometry

- B O'Neill, **Semi-Riemannian Geometry**, 1983
- S Lovett, **Differential Geometry of Manifolds 2nd Ed**, 2020

Note :

The hyperlinks in this lecture notes act as follows: The ■ hyperlink, when pressed, opens the relevant `pdf` in my (the Author's) local machine. This only works for my personal computer. The hyperlink on the **name of the book** links to a source of the book on the internet. Press that hyperlink to access that book.

Reminders

Exam Dates

- ☐ Long Exam 1 : **October 12, 2026, 9-12 AM**
- ☐ Long Exam 2 : **December 7, 2026, 9-12 AM**
- ☐ Oral Exam : **Finals week (TBA)**

Problem Set Dates

- ☐
- ☐
- ☐
- ☐
- ☐

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Chapter 1

Review of Newtonian Physics

1.1 Introduction

> “That gravity should be innate, inherent, and essential to matter, so that one body may act upon another at a distance through a vacuum, without the mediation of anything else... is to me so great an absurdity that I believe no man who has, in philosophical matters, any competent faculty of thinking can ever fall into it.” – Newton

We begin this course with the assumption that you have no idea what General Relativity is. How do we quantify what is gravity? What even is gravity? We begin with the classical equations of motion.

The classical equations of motion, first described by Galileo and formalized by Isaac Newton, have long served as the default model for describing the behavior of classical objects under the influence, or lack thereof, of external forces, and remain a fundamental part of physics education due to their success in describing matter within the scope of ordinary human observation. However, in the dawn of the 20th century, contradictions between these classical descriptions and experimental observations began to surface, making it apparent that a new description of reality was needed to reconcile theory with these results.

To motivate ourselves, we describe three concepts introduced over the history of classical mechanics and explain how it leads to certain contradictions or problems that motivated the realization and invention of general relativity

1.1.1 Newton's Law of Gravitation

Newton's Laws as he prescribed them is expressed as follows:

$$m_N \frac{d^2\mathbf{x}_N}{dt^2} = G \sum_M \frac{m_N m_M (\mathbf{x}_M - \mathbf{x}_N)}{\|\mathbf{x}_M - \mathbf{x}_N\|^3} \tag{1.1.1}$$

More often, it is shortened as:

Definition 1.1 : Newton's Law of Gravity

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \tag{1.1.2}$$

This is the familiar form of Newtonian gravity, and is perhaps one of the most recognizable equations in physics. This is what every high schooler knows and what every mentor presents, and is trivial to derive via experimentation.

We can separate the mass m experiencing the force from the mass M producing it, and write

$$\mathbf{F} = m_g\left(-\frac{GM}{r^2}\hat{\mathbf{r}}\right) \tag{1.1.3}$$

We can henceforth call m_g as the *gravitational mass* and the quantity $-\frac{GM}{r^2}\hat{\mathbf{r}}$ as the *gravitational field*

We can express it more generally by recognizing the force field is simply the negative gradient of a potential field.

$$\mathbf{F} = m_g(-\nabla\Phi) \tag{1.1.4}$$

where $\Phi = -\frac{GM}{r}$ is the gravitational potential produced by the source mass.

Thus, we can henceforth call m_g as the *gravitational charge.* It follows from the fact that

$$\text{charge} \times \text{field gradient} = force \tag{1.1.5}$$

This formulation is universal across all of physics.

Analogous to electromagnetism, where

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}} \tag{1.1.6}$$

Like the electric charge q, the gravitational charge m_g determines the strength of the interaction of the forces.

An important property that Newtonian gravitation has is that it has superposition. i.e., fields add together linearly. This is therefore also true of the gravitational potential.

For a system of point masses, the total potential field $\Phi(\mathbf{x})$ at a point $\mathbf{x}$ is given by the sum of the potentials produced by each individual mass:

$$\begin{aligned}\Phi(\mathbf{x}) &= \sum_i \Phi_i \\ &= -G\sum_i \frac{M_i}{|\mathbf{x}-\mathbf{r}_i|}\end{aligned}$$

We can extend this description from a discrete collection of point masses to a continuous mass distribution. Let the mass contained in an infinitesimal volume $d^3\mathbf{r}$ be

$$dm_i = \rho(t,\mathbf{r})\, d^3\mathbf{r} \tag{1.1.7}$$

Then we arrive with the continuum version of the gravitational potential field generated by a mass distribution.

$$\Phi(\mathbf{x}) = -G\int \frac{\rho(t,\mathbf{r})}{|\mathbf{x}-\mathbf{r}|}\, d^3\mathbf{r} \tag{1.1.8}$$

Here, $\mathbf{r}$ denotes the position of each source element of the mass distribution, while $\mathbf{x}$ denotes the point at which the potential is being evaluated.

We can now obtain a differential equation for the gravitational potential. Taking the Laplacian of Φ gives

$$\nabla^2\Phi = -G\int \rho(t,\mathbf{r})\, d^3\mathbf{r}\, \nabla^2\frac{1}{|\mathbf{x}-\mathbf{r}|} \tag{1.1.9}$$

Recall that this operation results to a dirac-delta function

$$\nabla^2 \frac{1}{|\mathbf{x}-\mathbf{r}|} = -4\pi\delta(\mathbf{x}-\mathbf{r}) \tag{1.1.10}$$

which relates the Laplacian of the inverse-distance potential to the Dirac delta function. Substituting this into the previous expression gives

$$\nabla^2\Phi = 4\pi G \int \rho(t,\mathbf{r})\, d^3\mathbf{r}\, \delta(\mathbf{x}-\mathbf{r}) \tag{1.1.11}$$

Recall the sifting property of the Dirac delta

$$\int f(r)\delta(x-r)\, dr = f(x) \tag{1.1.12}$$

From this we obtain

$$\nabla^2\Phi = 4\pi G\rho(t,\mathbf{x}) \tag{1.1.13}$$

Since $\mathbf{x}$ is an arbitrary point, we may relabel it as

$$\nabla^2\Phi(t,\mathbf{r}) = 4\pi G\rho(t,\mathbf{r}) \tag{1.1.14}$$

Thus, we arrived with the fundamental field equation which governs Newtonian Gravity

Theorem 1.1 : Field Equation for Newtonian Gravity

$$\nabla^2\Phi = 4\pi G\rho \tag{1.1.15}$$

The significance of a field equation is that it relates a field to its sources. In this case, the mass density ρ acts as the source of the gravitational potential field Φ.

Note the following

$$\Phi = -G\int \frac{\rho\, d^3\mathbf{r}}{|\mathbf{x}-\mathbf{r}|} \iff \nabla^2\Phi = 4\pi G\rho \tag{1.1.16}$$

The two are notionally equivalent. One is a differential equation and the other is the integral form. One can switch to and from them using the Laplacian and Green's functions[1]. It assigns the definition of the field, and fields assign values to every point in space.

[1]Green's functions are essentially the inverse of a laplacian, ∇^{-2}

However, this formulation presents a fundamental problem. Since the potential Φ is determined directly by the mass density ρ, any change in the mass distribution produces an immediate change in the gravitational field throughout space. There is no propagation delay or retardation. In other words, Newtonian gravity permits changes in the gravitational field to propagate instantaneously, which is incompatible with relativistic causality. This problem is one of the motivations for general relativity.

The corresponding requirement in GR is that, in the appropriate weak-field and slow-motion limit, general relativity must reduce to Newtonian gravity.

Now that we have the potential of the field that the source creates, and thus the force, how do we determine how an object moves in response to that field?

1.1.2 Newton's Equations of Motion

The quintessential equation governing the motion of classical bodies was expressed by Newton as

$$\mathbf{F} = m\mathbf{a} \tag{1.1.17}$$

Everybody and their mother knows this shit.

This states that if one sums of all the forces acting on an object, it must equal its mass multiplied by its acceleration:

$$\sum \mathbf{F} = m\mathbf{a} \tag{1.1.18}$$

We can rewrite the equation as

$$\mathbf{a} = \frac{1}{m}\sum \mathbf{F} \tag{1.1.19}$$

Notice that the m term acts as a kind of *resistance* term[2], similar to the concept of inertia. This is to be expected, since Newton's first law can essentially be understood as the statement that, in the absence of a net force, an object does not change its state of motion.

Hence, we name this quantity the *inertial mass*, m_i. Inertia denotes the difficulty or resistance of an object to a change in its state of motion.

[2] Recall Ohm's Law $V = IR$.

We can restate (1.1.19) with m replaced by m_i, to distinguish inertial mass from the gravitational mass introduced previously:

$$\mathbf{a} = \frac{1}{m_i}\mathbf{F} \tag{1.1.20}$$

Recall that we already have the definition of $\mathbf{F}$ from (1.1.4). Thus, to calculate the motion of an object under the influence of gravity, we simply calculate its acceleration:

$$\mathbf{a} = \frac{m_g}{m_i}(-\nabla\Phi) \tag{1.1.21}$$

Notice that this is similar in form to the equation of motion for a charged particle in an electric field.

$$\mathbf{a} = \frac{q}{m_i}\mathbf{E} \tag{1.1.22}$$

where m_i is the inertial mass, the measure of an object's resistance to changes in its motion. In both cases, the coupling to the field determines the force, while the inertial mass determines the resulting acceleration.

However, for calculations involving Newtonian gravity, the ratio of the gravitational charge and inertial mass $\frac{m_g}{m_i}$ is assumed to be 1. Thus, for gravity, the acceleration of a body is expressed as

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.23}$$

This assumes that

$$m_g = m_i, \tag{1.1.24}$$

This feels unnatural and unexpected, a truly remarkable coincidence in Newtonian theory. Why should it be the case that one force acts differently from other forces in terms of its charge carriers? Mass is a fundamental property that affects other forces, there's no immediate evidence why mass is gravity's charge carrier.

There is no *a priori* reason why the magnitude of gravitational force on a particle should equal the quantity that determines the particle's resistance to an applied force in general. They play fundamentally different roles in the theory. It is analogous to assuming that an object's coupling to some other field must be equal to its inertial mass, like an electron's electromagnetic properties being dependent on mass.

Nevertheless, for gravity this equality is accepted experimental fact without explanation.

1.1.3 Weak Equivalence Principle

The assumption that

$$m_i = m_g \tag{1.1.25}$$

is called the *weak equivalence principle.*

It follows that, for objects only affected by gravity, the equation of motion is simply

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.26}$$

That is, any test mass placed in the same gravitational field experiences the same acceleration, regardless of its mass or composition. The nature of the object itself does not appear in the equation of motion. The field prescribes the acceleration, the test body merely responds to it.

One could therefore remove the test object m and nothing about the gravitational field would change. In Newtonian mechanics, this is treated as a universal fact: the gravitational field exists independently of the particular body used to probe it.

However, this raises a deeper question.

One could argue that the stage for the motion of an object is the field itself. If we remove the object from some local region, the field remains.

By the same logic, if we remove the field itself, what remains? In Newtonian mechanics, space $\mathbb{R}^3$ remains as the stage on which the field and matter exist.

The influence of gravity has nothing to do with the nature of the test object itself. The gravitational field does not change when the test object is removed, and its motion depends only on the field at its location.

One might ask: if we remove all mass, would there not be no gravity? The answer is that we must distinguish the sources of the field from the test body. We are considering local interactions. Mass may exist somewhere outside the local region of interest and still generate a gravitational field within it. Given the local potential Φ, we can determine information about its sources, but the test body itself is not one of those sources.

Thus,

$$\mathbf{a} = -\nabla\Phi \tag{1.1.27}$$

is treated as inertial motion in a gravitational field.

In the stage analogy, one could think of $-\nabla\Phi$ as living on the stage of space $\mathbb{R}^3$. The field exists on space and determines how objects move through it. In some fields such as particle physics, that is the case, they treat gravity as a special field which acts on every particle and every particle responds to in the same way

However, in GR, $-\nabla\Phi$ is no longer treated as a field sitting on top of the stage. The gravitational field *is* the stage. The space time is what the rest of physics takes place in.

The strong equivalence principle extends this idea: locally, the laws of physics in a freely falling frame are the same as those of an inertial frame in the absence of gravity. In other words, locally, gravity can be transformed away by choosing an appropriate freely falling frame.

Inertial motion is therefore determined by the structure of space-time.

Physics takes place on the stage called spacetime, and GR governs that space time. The spacetime structure remains even after the particular body used to probe it is removed.

In GR, instead of the gravitational potential field Φ, we have the metric:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \tag{1.1.28}$$

The metric determines the geometry of spacetime and, consequently, the inertial trajectories of objects.

In particle physics, one generally begins with a fixed flat spacetime $\mathbb{R}^4$ and defines fields upon that background. Gravity can then be treated as a field on this background, rather than as the geometry of spacetime itself. This approach is sound as a field-theoretic formulation and provides a natural route toward quantizing gravity and describing gravitons, but it is ugly and inelegant. Regardless, you can bootstrap yourself from there and arrive on the EFE

The distinction of perspectives can be summarized as follows:

Metric

↙ ↘

Relativist perspective

Spacetime structure is variant
↓
Gravity is the manifestation of Riemannian spacetime curvature and it governs particle motion

Particle physicist perspective

Fixed spacetime, fields live on top of it
↓
Gravity is a special gauge field in $SO(3,1)$ which couples to everything universally

Regardless of the formalism, the different approaches must reproduce the same classical predictions. The geometric formulation of GR ultimately leads to the Einstein field equations.

1.1.4 A Theory of Gravity

General Relativity, to be a theory of gravity, must explain gravitational phenomena.

For example, Newtonian gravity became a theory of gravity because it explained the results Kepler had found. Kepler described the motion of the planets and discovered that their orbits were elliptical using data analysis, but he did not have an underlying physical theory explaining *why* they followed these trajectories.

Newton instead derived the observed behavior from first principles. His laws of motion and universal gravitation explained Kepler's laws and provided a general framework for celestial mechanics, and ended up predicting the whole of kinematics and dynamics.

As such, GR as a theory of gravity must be able to explain observed gravitational phenomena: relativistic corrections to celestial mechanics, gravitational waves, black holes, gravitational collapse, and, on the largest scales, cosmology.

Perhaps the grandest application of GR is cosmology. Newtonian gravity is fundamentally a local theory applied within a fixed space and time, while GR allows the spacetime itself to be dynamical. This makes it possible to formulate a theory describing the universe as a whole.

DISTINCTION: For something to be a theory of gravity, what must it possess?

1. **Dynamical variables**

   The theory must have variables describing the gravitational field in which objects move.

$$\begin{aligned} \text{Newton:} \quad & \Phi \\ \text{GR:} \quad & g_{\mu\nu} \end{aligned}$$

2. **A law describing how the field affects objects**

   The theory must describe how its dynamical variables determine the motion of objects.

   For Newtonian gravity:

$$\mathbf{a} = -\nabla\Phi. \tag{1.1.29}$$

   For GR, freely falling objects follow geodesics:

$$\frac{d^2x^\sigma}{d\lambda^2} + \Gamma^\sigma_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0, \tag{1.1.30}$$

   or, using the four-velocity u^μ,

$$u^\mu \nabla_\mu u^\nu = 0. \tag{1.1.31}$$

3. **A law describing how objects produce the dynamical variable**

   Finally, the theory must tell us how matter produces the gravitational field.

   For Newtonian gravity:

$$\nabla^2\Phi = 4\pi G\rho. \tag{1.1.32}$$

   For GR:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}. \tag{1.1.33}$$

   Since the two formalisms attempt to explain the same phenomenon, we demand that the GR formulation reduces to the Newtonian formulation in weak fields.

Remark :

Notice that $\nabla^2\Phi = 4\pi G\rho$ and $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu}$ are inherently similar. They basically describe the same thing, but in different fields of geometry. Note the following:

$$\underbrace{\nabla^2\Phi}_{\text{some combination of } \partial^2} = \underbrace{4\pi G\rho}_{\text{constant multiplied to matter distribution}}$$

$$\underbrace{R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R}_{\text{some combination of } \partial^2} = \underbrace{8\pi GT_{\mu\nu}}_{\text{constant multiplied by matter distribution}}$$

These three ingredients are the basic structure of a theory of gravity: a dynamical variable, a law governing how it affects matter, and a field equation governing how matter produces it.

GR is inherently nonlinear. Going from the field to the motion of objects is already more complicated than in Newtonian gravity, but the field equation itself is also nonlinear: the metric determines the curvature, while the curvature determines the metric.

Solving the Einstein field equations for realistic systems is therefore often extremely difficult. In many cases, exact solutions cannot be obtained, and the equations must be solved numerically. Careers have been made doing this. This is the domain of *numerical relativity*.

In the next chapters, we will explore how gravity can be explained by general relativity.

1.2 Einstein's Equivalence Principle

1.2.1 Inertial Frames

The Weinberg definition is that, at every point in spacetime, i.e. at every (t, x, y, z) in a gravitational field, it is possible to choose a locally inertial frame, i.e. a freely falling coordinate system, such that the "laws of nature" take the same form as they do in an unaccelerated Cartesian coordinate system in the absence of gravity.

In other words, fields that are accelerating are locally indistinguishable from gravity. This is the basic idea behind the equivalence principle.

Gravity can therefore be regarded as a fictitious force, because it can be removed by a coordinate transformation. It is there only because we are using the wrong coordinate system.

The equivalence principle states that it is always possible to remove gravity locally. Therefore, gravity is a fictitious force" that can be "removed" when describing the local field in an experiment.

The important difference between gravity and other fictitious forces is that gravity can only be removed locally, while other fictitious forces can be removed globally. The reason is that a gravitational field has spatial variation. Even after transforming away the gravitational acceleration at one point, neighboring points may experience slightly different accelerations.

We say that, regardless of the mass of the object, it does not take part in determining its own gravitational acceleration. The gravitational acceleration is universal.

The measurement of differences between these gravitational fields is called *tidal* measurement, or simply the measurement of tides. These tidal effects cannot be transformed away over an extended region.

1.2.2 Universality of Free Fall

The usual way this is expressed is to imagine a person in a closed box with no reference to the outside world. To gain information about his environment, he begins conducting a free-fall experiment. Suppose they perform some experiment in which they can measure the acceleration of objects.

For an object o_1, they measure

$$\mathbf{a}_1 = \frac{m_{1g}}{m_{1i}}\mathbf{g} \tag{1.2.1}$$

and for a second object o_2, they can measure

$$\mathbf{a}_2 = \frac{m_{2g}}{m_{2i}}\mathbf{g}. \tag{1.2.2}$$

The universality of free fall is the statement that

$$\mathbf{a}_1 = \mathbf{a}_2 \tag{1.2.3}$$

under the same circumstances, regardless of the nature or mass of the objects. This is called the *universality of free fall.*

Suppose that the box is accelerating upward with acceleration $g\hat{\mathbf{z}}$.

$$\mathbf{a}_{\text{box}} = g\hat{\mathbf{z}} \tag{1.2.4}$$

The observer releases the ball. From his perspective, the ball appears to accelerate downwards.

$$\mathbf{a}_{\text{ball}} = -g\hat{\mathbf{z}} \tag{1.2.5}$$

Therefore, it is reasonable to conclude that "there is a gravitational field pulling the force downward".

But there isn't. The ball is actually just continuing along its inertial trajectory while the box accelerates upward.

Now compare this with another box which is stationary, with another observer sitting on the surface of Earth.

The observer releases the same ball, and it falls downward with approximately

$$\mathbf{a}_{\text{ball}} = -g\hat{\mathbf{z}} \tag{1.2.6}$$

Inside the box, the experiment looks the same. So locally, motion due to acceleration is indistiguishable from motion incduced by a gravitational field. There is no experiment that can be performed by the two observers that can differentiate one from another. This is the strong equivalence principle.

In Newtonian language, this corresponds to the equality of gravitational and inertial mass. The important point for the equivalence principle is that all sufficiently small freely falling objects respond to the gravitational field in the same way.

Now, imagine the observer is in a box or elevator that is freely falling towards Earth. The box and everything inside it are accelerating downward together. Suppose the observer release a ball inside the box. The ball is falling toward Earth, but so is the box. Since both have essentially the same gravitational acceleration,

$$\mathbf{a}_{\text{ball}} \approx \mathbf{a}_{\text{box}} \tag{1.2.7}$$

their relative acceleration is approximately zero:

$$\mathbf{a}_{\text{relative}} = \mathbf{a}_{\text{ball}} - \mathbf{a}_{\text{box}} \approx 0 \tag{1.2.8}$$

so the ball appears to float. The observer could throw it, and it would move approximately in a straight line at constant velocity relative to the observer.

The observer would therefore reasonably conclude "There is no gravity here. I'm in an inertial frame."

But the observer actually is in Earth's gravitational field. In actual inertial motion, far from any gravitating bodies, the same observations will take place. Therefore, a falling observer is indistinguishable from an inertial observer. That is the weak equivalence principle.

In a simple example, we can write the gravitational field as a non-inertial frame:

$$\ddot{y} = -g \tag{1.2.9}$$

We can remove gravity, however, via certain transformations. Let

$$z = y - \frac{1}{2}gt^2 \tag{1.2.10}$$

This results in

$$\ddot{z} = 0. \tag{1.2.11}$$

This is an inertial coordinate system, since there is no perceived acceleration. The point is that gravity can be removed locally by choosing an appropriate freely falling coordinate system.

The person in the box cannot tell whether they are accelerating upward with constant acceleration g, or whether they are freely falling under the influence of gravity. In a sufficiently small region, the two situations are experimentally indistinguishable.

Gravity is therefore, locally, simply the manifestation of being in the wrong frame.

Inertial means the absence of gravitational acceleration in the local frame. More precisely, an inertial frame is one in which freely falling objects move without coordinate acceleration.

Weak, Strong, and Einstein Equivalence Principles We have the weak equivalence principle, where the "laws of nature" are understood primarily as the laws of mechanics and the universality of free fall.

For the strong equivalence principle, one extends the statement to gravitationally self-interacting or strongly gravitating systems, so that the equivalence applies more broadly than to test particles alone.

The Einstein equivalence principle goes further: "laws of nature" genuinely means all laws of nature. It is one of the conceptual foundations of general relativity.

In any theory of gravity, one can attempt to remove g locally by an appropriate choice of coordinates. What makes general relativity special is the geometric interpretation of what remains after the local gravitational acceleration has been removed.

It means that gravity is genuinely fictitious locally if no experiment confined to that sufficiently small region can tell that we are in a gravitational field. If one can remove gravity, then what remains should be the stage itself: the underlying spacetime structure and its geometry.

In summary, over a sufficiently small region of space and time, a fictitious gravitational field produced by acceleration is indistinguishable from a gravitational field produced by mass (strong equivalence principle). Likewise, within a sufficiently small region of space, a freely falling observer cannot distinguish between free fall in a gravitational field and inertial motion in the absence of gravity (weak equivalence principle). Therefore, a freely falling observer is locally an inertial observer.

1.3 History of Relativity

The notion of inertiality plays a major role in relativity. Let us first explain relativity in its historical context.

We begin with special relativity and Newtonian gravity. The combination of these ideas eventually leads us toward general relativity, where gravity is incorporated into the structure of spacetime itself.

1.3.1 Relativity before Einstein

What is relativity? It is an old notion that existed even by the time of Newton. The essential idea is that there should exist observers for which the laws of nature take the same form. Regardless of where the observer is or how they move within the appropriate class of frames, the laws of nature should be invariant.

This is the equivalence of *inertial observers*: the laws of nature take the same form for all inertial observers.

Aristotle did not have this notion. He believed that there was a special place toward which objects naturally moved: the center of the universe, identified in practice with the center of the Earth. One could therefore imagine that this special region defined the natural frame in which the laws of nature took their proper form.

It made sense within that picture to express physics relative to the center of the Earth, since all other frames could be considered as moving relative to this preferred location. For example, one could imagine waves in a pond as being fixed relative to the pond itself, with the pond providing a preferred rest frame.

Galileo rejected this notion by positing that there is not a single preferred inertial state. Instead, there is a class of observers for whom the laws of mechanics are equally valid. These observers move uniformly relative to one another.

This is called *Galilean relativity.* There is a class of observers which all move relative to each other with uniform velocities.

The transformation between two such frames can be written as

$$t' = t \tag{1.3.1}$$

$$\mathbf{x}' = \mathbf{x} - \mathbf{v}t \tag{1.3.2}$$

This relation explains how one can switch to the frame of reference of a second observer moving at velocity $\mathbf{v}$ relative to the first.

In that case, one has the following invariance of Newtonian mechanics:

$$\mathbf{F} = m\mathbf{a} \implies \mathbf{F}' = m\mathbf{a}'. \tag{1.3.3}$$

Newtonian theory therefore satisfies Galilean relativity: it is invariant under Galilean transformations.

From the development of Newtonian mechanics onward, classical mechanics became the fundamental framework of physics. From 1687 onwards, the only fundamental laws of physics is Newtonian physics.

1.3.2 Maxwell's Results

For roughly three centuries, Newtonian mechanics was the jewel of physics. Then electromagnetism broke this picture.

In 1862, Maxwell showed that Maxwell's equations are not Galilean invariant. They therefore do not satisfy Galilean relativity in the same way that Newton's equations do. This challenged centuries of orthodoxy.

The response of academia was initially to deny that Maxwell's equations were fundamental in the same sense as Newtonian mechanics. One possibility was that Maxwell's equations were only correct in the rest frame of some physical medium, and that medium was called the *aether*.

The aether was imagined as an invented medium through which electromagnetic waves propagated. In this picture, Maxwell's equations described electromagnetic waves in the same broad conceptual sense that waves in a pond are described relative to the pond.

Light would then move at speed c only in the rest frame of the aether.

This led naturally to attempts to detect the Earth's motion through the aether. The Michelson–Morley experiment was a null experiment: the expected difference in the measured speed of light was not observed, and the result indicated that c was the same in the different experimental directions.

People fought against this result, as they should have. The natural response to an experimental discrepancy is to try to preserve a successful theory if possible and search for a physical explanation for the new result.

Ad Hoc Solutions to the Aether Problem. One proposed solution was Lorentz contraction. Objects would contract in the direction of motion through the aether, although there was initially no satisfactory physical explanation for why they should do so. It was essentially an ad hoc solution introduced to maintain consistency with the null result.

Another proposed idea was *ether dragging*. Imagine that the ether is not a fixed frame, but behaves something like molasses or syrup, so that the Earth in motion drags the ether around it. The Earth would then remain locally at rest with respect to the ether, and there would consequently be no observed change in c.

The debates went on for decades. People had built careers around various versions of these theories, and many ad hoc solutions were introduced to preserve the compatibility of Maxwell's theory with Galilean relativity.

There were dozens of theories introduced in the hope of making Maxwell's equations consistent with the Galilean framework.

1.3.3 Einstein's Contribution

Einstein came in 1905 with his *annus mirabilis* papers.

The three major papers associated with this period gave rise to major developments in quantum theory, statistical mechanics, and special relativity. Einstein developed these ideas all before turning 24.

Everyone had been focused on finding a way to mesh Galilean relativity with Maxwell's equations. The common strategy was to deny the fundamentality of Maxwell's equations by claiming that they only worked in the preferred frame of the aether.

Einstein instead reversed the argument.

Einstein posited that Maxwell's equations are fundamental, while Newtonian mechanics is not fundamental. Newtonian mechanics was a theory that had been successful for three centuries, but its enormous historical success did not make it fundamentally immune to revision.

His solution is:

- adopt c as universal;
- accept Maxwell's equations as fundamental.

As such, the equations governing Galilean relativity must be dropped so that c remains constant. By consequence, one also has to give up or modify Newtonian mechanics and stop treating it as fundamentally exact.

One has to divorce oneself from the notion that Newtonian mechanics is fundamental.

The notion of universal observers is still true: there is a class of observers for whom the laws of motion retain the same form. However, to reconcile this with the invariance of c, one must change the transformation law between observers.

This leads to the unification of space and time and to the Lorentz transformations.

Lorentz Transformations and Covariance Maxwell's theory is relativistic. It is relativistically covariant: under the appropriate relativistic transformations, its mathematical form does not change.

The task is therefore to find the correct transformation laws that keep Maxwell's equations invariant.

The Poincaré-Lorentz transformations provide the appropriate transformations:

$$ct' = \gamma(ct - \beta x) \tag{1.3.4}$$

$$x' = \gamma(x - \beta ct) \tag{1.3.5}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \tag{1.3.6}$$

$$\beta = \frac{v}{c}. \tag{1.3.7}$$

The project, according to Einstein, is to rewrite all of physics so that it is in "relativistically covariant" form: the equations should be invariant in form from one inertial observer to another[3].

[3] This isn't too difficult back in 1905, since the only fundamental theories of physics at the time were Newton's and Maxwells.

The form of these transformations is

$$x^{\mu'} = \Lambda^{\mu}{}_{\nu} x^{\nu}. \tag{1.3.8}$$

Where

$$x^{\mu} = (ct, x^1, x^2, x^3) \tag{1.3.9}$$

and

$$\Lambda^{\mu}{}_{\nu} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.3.10}$$

Now, the force equation becomes

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} \implies \frac{dp^{\mu}}{d\tau} = F^{\mu}. \tag{1.3.11}$$

The second equation reduces to the first in the appropriate nonrelativistic, low-velocity limit.

How About Gravity? Gravity is simply

$$\nabla^2 \Phi = 4\pi G \rho. \tag{1.3.12}$$

This is not relativistically covariant, since it is written as a theory involving only space and does not incorporate space and time into a unified relativistic structure.

One can try to modify it so that

$$\Box \Phi = 4\pi G T^{\mu}{}_{\mu}. \tag{1.3.13}$$

This is known as Nordström gravity. But this theory is incorrect as a theory of gravity because it does not predict the bending of light.

However, Einstein was bothered by the equivalence principle. Why is there an equivalence of falling? Why does gravity have this peculiar property that all objects fall in the same way?

This question leads to the deeper geometric interpretation of gravity.

Minkowski Spacetime It was Minkowski in 1907 who placed these transformations in their modern geometric form.

One can make sense of special relativity by imagining space and time as a single structure: *spacetime*. This spacetime possesses a metric, now known as the Minkowski metric,

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.3.14}$$

and the transformations that preserve the Minkowski metric are the correct relativistic transformations.

The key conceptual step is that space and time are no longer treated as two completely separate structures. They form a single spacetime geometry, and inertial observers correspond to the natural geometric structure of this spacetime.

1.3.4 From Special to General Relativity

In 1915, Einstein formalized his theory of General Relativity, where he formalized the notion of spacetime and recognized that spacetime need not be described only by the Minkowski metric.

The replacement is

$$\eta_{\mu\nu} \to g_{\mu\nu}. \tag{1.3.15}$$

The Minkowski metric is therefore promoted from the fixed metric of special relativity to the more general spacetime metric $g_{\mu\nu}$, which may vary from place to place.

Einstein expanded the notion of inertial observers to all observers in free fall within spacetime. Inertial observers are therefore no longer merely observers moving with uniform velocity relative to one another. They are also freely falling observers, following geodesics of the spacetime geometry.

He got the basic idea relatively early, but obtaining the field equations took much longer.

Relativity is very, very old, and the important thing Einstein did was expand it into a general theory in which the geometric structure itself becomes dynamical.

Minkowski's relativity is special relativity, represented by η, called special because it is restricted to a class of observers moving uniformly relative to one another.

The general form, represented by g, becomes general relativity, with gravity incorporated into the geometry of spacetime.

Machian and Relational Ideas There is a distinction here involving the idea of being *Machian*. A Machian picture suggests that everything should be relational and that nothing should be absolute. In essense, Einstein was a Machian, but the theory that he founded is fundamentally not Machian.

General relativity has a deeply relational character, although it is not simply Machian in every possible sense. The geometry of spacetime itself becomes part of the physical description rather than serving merely as an immutable background stage.

Relativity is more general than Light There is a common tendency to regard light as being intrinsically tied to relativity. Historically, however, light played a more specific role. Even if Maxwell's theory were somehow removed or replaced by another theory, the principle of relativity would remain.[4]

Light has no fundamental role in the concept of relativity itself. Its importance is primarily historical: Maxwell's equations revealed that electromagnetic waves propagate at a universal speed that is incompatible with Galilean kinematics. This compelled physicists to reconsider the Galilean transformation and ultimately led to the Lorentz transformation. The principle of relativity, however, is conceptually distinct from electromagnetism.

Relativity therefore is a metaprinciple. It is a more general principle concerning the relationship between physical laws and different

[4] Sir here goes on a rant about news sites which sensationalize 'slowing down light speed' experiments as 'putting to question relativity'

observers. Maxwell's theory inspired and helped reveal the structure of spacetime, but it is not the conceptual foundation of the principle of relativity itself.

Light is therefore not intrinsically tied to relativity. The relativistic structure of spacetime can be developed without taking electromagnetism or light as its starting point. Likewise, general relativity can be formulated without making light fundamental to its derivation, while still arriving at the same gravitational field equations.

The next lecture will take precisely this approach: we will develop relativity without making any mention of light and arrive at the same equations that describe gravity in Einstein's theory.

The Role of c The essential role of Maxwell's theory in the historical development of relativity is to identify a universal invariant speed, c, that characterizes the structure of spacetime. However, the existence of such an invariant speed does not depend specifically on electromagnetism. We can construct relativistic kinematics with any finite, nonzero value of the invariant speed.

Consider the limiting cases. If we take

$$c \to \infty,$$

we recover Galilean relativity.[5] In this limit, the effects associated with the relativity of simultaneity vanish, and time becomes absolute.

If instead we take

$$c \to 0,$$

we obtain Carrollian relativity,[6] the ultra-relativistic counterpart of the Galilean limit. In this limit, the causal structure becomes degenerate in such a way that spatial motion relative to an inertial observer is effectively frozen.

If instead we asign c to be any fixed finite value,

$$c \in (0, \infty),$$

we obtain the Lorentzian kinematic structure underlying special relativity and, ultimately, general relativity. The particular numerical value of

[5] This was in fact the case during the time of Galileo. It was a mainstream hypothesis that the speed of light was infinite

[6] This is in reference to Lewis Carrol's *Alice in Wonderland*, likening the bizarre events that happen in such conditions to the plot of said work

c is therefore not essential to the structure of the theory; any fixed finite nonzero value could serve as the invariant speed, with the numerical value depending on the choice of units and on the physical theory being described.

The constant c is commonly called the *speed of light* because electromagnetic radiation propagates at this invariant speed. Historically, the value of c emerged from Maxwell's theory as the propagation speed of electromagnetic waves. Modern physics, however, understands c more fundamentally as a property of spacetime itself: light travels at c because photons are massless, just as gravitational waves propagate at c. Massive particles cannot reach c, although they may approach it arbitrarily closely.

Thus, Maxwell's theory is not required in order to derive the existence of a finite invariant speed. Rather, it was the first physical theory to reveal that such a speed exists in nature and to give it the numerical value

$$c = 299\,792\,458\,\mathrm{ms}^{-1}$$

We can therefore develop a theory of relativity without making any reference to light at all. Starting from the principles of spacetime symmetry, we will see that a finite invariant speed can arise from the kinematic structure itself. We will develop this approach in the next chapter.

Chapter 2

Special Relativity

2.1 Relativity without Light

We begin this section by doing a derivation of special relativity in large contrast with standard methods which introduce relatvity with light-based thought experiments. Our derivation follows the elegant logic developed by von Ignatowsky (1910) and later refined by others, showing that the Lorentz transformations follow from fundamental symmetries alone.

2.1.1 The Transformation Problem

Consider two inertial reference frames S and S', where S is taken as a *rest* frame of reference, and S' moves with a constant velocity v along the x axis with respect to S.

images/frames_in_motion.pdf

In Galilean transformations, the relationship betwen the two reference frames can be described as

$$\begin{aligned} x' &= x + vt \\ t' &= t' \end{aligned}$$

That is, expressed in matrix form,

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} 1 & v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.1}$$

This relationship is what is naively assumed during day to day life, and in fact, it is readily apparent for observers without any knowledge of relativity. However, it has some issues. It implies that time t is absolute, and, the reference frames S and S' are not strictly equivalent.

As such, we need a more robust transformation. We seek the most general transformation between the coordinates of the two frames.

$$x' = \overline{X}(x, t; v) \tag{2.1.2}$$

$$t' = \overline{T}(x, t; v) \tag{2.1.3}$$

The functions $\overline{X}$ and $\overline{T}$ are transformation functions that are initially unknown and undetermined.

Our task, then, is to determine them from general physical principles. Simply logic and mathematics, without any reference to experimental data whatsoever. This is, essentially, a purely mathematical approach. Eventually, our goal is to obtain a transformation in the form of Lorentzian transfromations.

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \gamma \begin{bmatrix} 1 & -v \\ -\beta & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.4}$$

where $\gamma = \dfrac{1}{\sqrt{1-\beta^2}}$ and $\beta = \dfrac{v^2}{c^2}$

An interesting question arises: Where does c come from if we never assume light?

2.1.2 Principle of relativity

The first assumption we make is the principle of relativity.

> The laws of physics take the same form in all inertial reference frames.

In particular, S and S' are physically equivalent. There is no reason to regard the transformation from S to S' as fundamentally different from the transformation from S' back to S.

If S' moves at velocity v relative to S, then S moves at a velocity $-v$ relative to S'. Hence, the inverse transformation of (2.1.3) must have the same functional form:

$$\begin{aligned} x' &= \overline{X}(x, t; -v) \\ t' &= \overline{T}(x, t; -v) \end{aligned}$$

This implies that if we apply the forward transformation followed by the inverse transformation, we recover the original coordinates:

$$\begin{aligned} x &= \overline{X}\big(\overline{X}(x,t,v), \overline{T}(x,t,v), -v\big) \\ t &= \overline{T}\big(\overline{X}(x,t,v), \overline{T}(x,t,v), -v\big) \end{aligned} \tag{2.1.5}$$

2.1.3 Isotropy of Space

The isotropy of space means that physics has no preferred direction. In particular, the physics cannot distinguish between the $+x$ and $-x$ directions. This imposes the following:

Consider reversing the spatial axis,

$$x \to -x. \tag{2.1.6}$$

Under this reversal, the relative velocity also changes sign,

$$v \to -v. \tag{2.1.7}$$

The spatial coordinate must therefore transform as

$$x' = -\overline{X}(x, t; v), \tag{2.1.8}$$

while time, being a scalar under spatial reflection, satisfies

$$t' = \overline{T}(x, t; v). \tag{2.1.9}$$

This distinction is important: x changes sign under spatial reflection, but t does not.

Under time reversal combined with spatial reflection:

$$\begin{aligned} x' &= -\overline{X}(-x, t, -v) \\ t' &= \overline{T}(-x, t, -v) \end{aligned} \tag{2.1.10}$$

For these to be consistent with the original transformations, we require:

$$\begin{aligned} \overline{X}(-x, t, -v) &= -\overline{X}(x, t, v) \\ \overline{T}(-x, t, -v) &= \overline{T}(x, t, v) \end{aligned} \tag{2.1.11}$$

2.1.4 The Invariance of Straigt Lines

Free particles must move along straight lines in both frames. This is required by Newton's first law, the principle of inertia. In space time coodinates, we denote $x^\mu = (x, t)$[7] and $x'^\mu = (x', t')$. Straight line motion in S is therefore characterized by:

$$\frac{d^2 x^\mu}{d\tau^2} = 0 \tag{2.1.12}$$

Where τ is some affine parameter[8].

The principle of inertia demands that straight lines map to straight lines.

$$\frac{d^2 \overline{x}^\mu}{d\tau^2} = 0 \implies \frac{d^2 x'^\mu}{d\tau^2} = 0 \tag{2.1.13}$$

[7] This is tensor notation, where μ identifies the component. In this 1+1 dimensional case, $x^1 = x$ and $x^0 = t$. However, in general, $x^\mu = \left[t, x, y, z\right]^\top$

[8] Affine parameter is an arbitrary parameter for which the geodesic equation is said to hold and takes the form of uniform motion wtihout introducing artificial acceleration through the choice of parameter. i.e, affine parameters allow reparametrization to be linear transformations. I don't understand it either, go figure. In this case, τ is called the *proper time*

Using the chain rule, we can more explicity show that the derivative is a product of a jacobian term and a velocity term

$$\frac{dx'^{\mu}}{d\tau} = \sum_{\nu} \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{dx^{\nu}}{d\tau} \tag{2.1.14}$$

Here, we impose the Einstein summation convention, where twice repeated indices are implicitly summed. Explicitly:

$$\frac{dx'^{\mu}}{d\tau} = \frac{\partial \overline{X}^{\mu}}{\partial x^{0}} \frac{dx^{0}}{d\tau} + \frac{\partial \overline{X}^{\mu}}{\partial x^{1}} \frac{dx^{1}}{d\tau} \tag{2.1.15}$$

Henceforth, we write this compactly as:

$$\frac{dx'^{\mu}}{d\tau} = \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{dx^{\nu}}{d\tau} \tag{2.1.16}$$

with the understanding that ν is summed over its range.

Continuing the derivation, we have

$$\frac{d^2 x'^{\mu}}{d\tau^2} = \frac{d}{d\tau}\left(\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}}\right) \frac{dx^{\nu}}{d\tau} + \frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{d^2 x^{\nu}}{d\tau^2} \tag{2.1.17}$$

The second term vanishes by the straight-line condition in S

$$\cancel{\frac{d^2 x'^{\mu}}{d\tau^2}} = \frac{d}{d\tau}\left(\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}}\right) \frac{dx^{\nu}}{d\tau} + \cancel{\frac{\partial \overline{X}^{\mu}}{\partial x^{\nu}} \frac{d^2 x^{\nu}}{d\tau^2}} \tag{2.1.18}$$

Therefore, from this result, we demand that the following relation must hold.

$$0 = \frac{\partial^2 \overline{X}^{\mu}}{\partial x^{\nu}\, \partial x^{\sigma}} \frac{dx^{\nu}}{d\tau} \frac{dx^{\sigma}}{d\tau} \tag{2.1.19}$$

For arbitrary velocities $\dfrac{dx^{\nu}}{d\tau}$, this requires:

$$\frac{\partial^2 \overline{X}^{\mu}}{\partial x^{\nu}\, \partial x^{\sigma}} = 0 \tag{2.1.20}$$

Therefore, $\overline{X}^{\mu}$ must be a linear function of the coordinates. Hence, we can write:

$$\overline{X}(x, t, v) = A(v)x + B(v)t + E(v)$$
$$\overline{T}(x, t, v) = C(v)x + D(v)t + F(v)$$

These are just equations of general straight lines.

2.1.5 Homogeneity of Spacetime

The next assumption we use is the homogeneity of space time. Spacetime homogeniety implies no preferred origin.

Any point in both S and S' can be an origin and spacetime would remain the same, so therefore we can choose the location of the origin for both.

Therefore, we demand that the origin of S gets mapped to the origin of S' at $t = 0$.

$$x = t = 0 \iff x' = t' = 0 \tag{2.1.21}$$

This is not really essential. However, it makes the derivation simpler and easier, since we can then set the constants $E(v)$ and $F(v)$ to zero. Doing otherwise would make the constant terms indeterminable, which would increase the difficulty od the problem. However, the general result is the same. With this assumption, we can reduce our transformations to

$$\begin{aligned} \overline{X}(x,t,v) &= A(v)x + B(v)t \\ \overline{T}(x,t,v) &= C(v)x + D(v)t \end{aligned} \tag{2.1.22}$$

Thus, only four free functions of velocity remain to be determined: $A(v), B(v), C(v)$, and $D(v)$. Our task, then, is to reduce these four free functions to just one function by exploiting properties that were previosuly shown.

2.1.6 Parity Constraints

Applying the parity conditions from Eq. (2.1.11) to the linear forms, we have

$$\begin{aligned} \overline{X}(-x,t,-v) &= -\overline{X}(x,t,v) \\ \overline{T}(-x,t,-v) &= \overline{T}(x,t,v) \end{aligned} \tag{2.1.23}$$

This yields the following:

$$\begin{aligned} -A(-v)x + B(-v)t &= -A(v)x - B(v)t \\ -C(-v)x + D(-v)t &= C(v)x + D(v)t \end{aligned} \tag{2.1.24}$$

For these to hold for all x and t, the following must be true

$$\begin{aligned} A(-v) &= A(v) \\ D(-v) &= D(v) \\ B(-v) &= -B(v) \\ C(-v) &= -C(v) \end{aligned} \tag{2.1.25}$$

Thus, $A(v)$, $D(v)$ are even functions of v, and $B(v)$, $C(v)$ are odd functions of v

2.1.7 Inverse transformation and composition constraints

The inverse transformation from S' to S is

$$\begin{bmatrix} x \\ t \end{bmatrix} = \begin{bmatrix} A(-v) & B(-v) \\ C(-v) & D(-v) \end{bmatrix} \begin{bmatrix} x' \\ t' \end{bmatrix} \tag{2.1.26}$$

Explicitly,

$$\begin{aligned} x &= A(-v)x' + B(-v)t' \\ t &= C(-v)x' + D(-v)t' \end{aligned} \tag{2.1.27}$$

Using the parity relations (2.1.25),

$$\begin{aligned} x &= A(v)x' - B(v)t' \\ t &= -C(v)x' + D(v)t' \end{aligned}$$

Substituting

$$\begin{aligned} x' &= A(v)x + B(v)t \\ t' &= C(v)x + D(v)t \end{aligned}$$

to (2.1.27), we obtain

$$\begin{aligned} x &= \quad A(Ax + Bt) - B(Cx + Dt) \\ t &= -C(Ax + Bt) + D(Cx + Dt) \end{aligned}$$

Collecting coefficients we have,

$$\begin{aligned} x &= (A^2 - BC)x + B(A - D)t \\ t &= C(D - A)x + (D^2 - BC)t \end{aligned}$$

Since x and t are arbitrary, the coefficients must satisfy

$$\begin{aligned} A^2 - BC &= 1 \\ B(A-D) &= 0 \\ C(D-A) &= 0 \\ D^2 - BC &= 1 \end{aligned} \tag{2.1.28}$$

So far, nothing specifically relativistic has appeared, there's no magic yet. We just used natural assumptions that come from geometry, and principle of relativity.

2.1.8 Two possibilities

The conditions

$$B(A-D) = 0 \tag{2.1.29}$$

$$C(D-A) = 0 \tag{2.1.30}$$

give us two cases.

Case 1: $A \neq D$

If $A \neq D$, then, we are forced to conclude

$$B = C = 0 \tag{2.1.31}$$

The remaining equations give

$$A^2 = D^2 = 1 \tag{2.1.32}$$

Therefore,

$$\begin{aligned} A &= \pm 1 \\ D &= \mp 1 \end{aligned}$$

Thus, the transformation becomes

$$\begin{aligned} x' &= \pm x \\ t' &= \mp t \end{aligned}$$

This is a trivial transformation, corresponding to an identity/reflection type transformation rather than a continuosly varying boost between each reference frame. This is obvious. Of course, when you flip a frame, all the assumptions and constraints still hold. This is uninteresting.

They are therefore not the nontrivial transformation we seek.

Case 2: $A = D$

This is the nontrivial condition. The physically interesting case is therefore

$$A(v) = D(v). \tag{2.1.33}$$

The condition

$$A^2 - BC = 1 \tag{2.1.34}$$

then gives

$$BC = A^2 - 1. \tag{2.1.35}$$

provided that $B \neq 0$ Thus:

$$C = \frac{A(v)^2 - 1}{B(v)}. \tag{2.1.36}$$

We have reduced the four unknown free functions to two, since $A = D$, and once you know A and B, you know C.

2.1.9 The motion of the origin of S'

Consider the motion of S' through S. The spatial origin of (S') is defined by

$$x' = 0 \tag{2.1.37}$$

Since S' moves at velocity v relative to S, its origin follows the trajectory

$$x = vt \tag{2.1.38}$$

But every event on this worldline must satisfy $x' = 0$. Again, that is for the spatial origin. That means that entire *world* line needs to be mapped to $x' = 0$.

Using

$$x' = Ax + Bt \tag{2.1.39}$$

we therefore have

$$\begin{aligned} 0 &= A(vt) + Bt \\ &= (Av + B)t \end{aligned}$$

Since this must hold for an arbitrary t, then

$$B(v) = -vA(v) \tag{2.1.40}$$

Now we only have one free function, $A(v)$.

Consequently, we have

$$C = \frac{A^2(v) - 1}{vA(v)} \tag{2.1.41}$$

Thus, we have reduced the transformation to the following

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} A & -vA \\ -\dfrac{(A^2-1)}{vA} & A \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.42}$$

Now we only need to know A, except we dont know what properties of A is. The only thing we can say for now is that matrix needs to reduce to the identiy matrix when $v = 0$

Thus we impose

$$A(v) \to 1, \quad v \to 0.$$

Since the 1,0 term is a fraction with v in the denominator, it must vanish sufficiently fast. therefore, $A^2 - 1 \approx \mathcal{O}\big(v^2\big)$

2.1.10 Introduction of a third observer

Consider a third inertial frame S''.

Suppose S'' moves with constant velocity u relative to S', while S' moves with constant velocity v relative to S.

It's the same framework, then. The transformation from S' to S'' takes a similar form as from S to S'. Let

$$M(v) = \begin{bmatrix} A(v) & -vA(v) \\ -\dfrac{(A(v)^2-1)}{vA(v)} & A(v) \end{bmatrix} \tag{2.1.43}$$

We can therefore denote the transformation from the S frame to the S' frame to be

$$x'^{\mu} = M(v)x^{\mu} \tag{2.1.44}$$

Since the transformation is the same, then the transformation from S' to S'' is

$$x''^{\mu} = M(u)x'^{\mu} \tag{2.1.45}$$

Suppose that S'' is moving at a speed w relative to S. Then

$$x''^{\mu} = M(w)x^{\mu} \tag{2.1.46}$$

However, we demand the closure of transformations under composition. i.e, the form of the transformation is the same, just with a different argument. Then, the transformation from S to S'' is obtained by composing the two transformations from the intermediate reference frame.

$$x''^{\mu} = M(u)M(v)x^{\mu} \tag{2.1.47}$$

That is, the form of the transformation from S to S'' is the same form as the transformations from the other frames of reference.

$$M(w) = M(u)M(v) \tag{2.1.48}$$

The crucial requirement is that the composition of two allowed transformations must itself be an allowed transformation.

2.1.11 The Emergence of an Invariant Speed

We evaluate the composition by doing matrix multiplication.

$$M(w) = \begin{bmatrix} A(u) & -uA(u) \\ -\dfrac{(A(u)^2-1)}{uA(u)} & A(u) \end{bmatrix} \begin{bmatrix} A(v) & -vA(v) \\ -\dfrac{(A(v)^2-1)}{vA(v)} & A(v) \end{bmatrix}$$

Therefore, we have

$$M(w) = \begin{bmatrix} A(u)A(v) + \dfrac{uA(u)\big(A(v)^2-1\big)}{vA(v)} & -(u+v)A(u)A(v) \\ -\dfrac{\big(A(u)^2-1\big)A(v)}{uA(u)} - \dfrac{A(u)\big(A(v)^2-1\big)}{vA(v)} & A(u)A(v) + \dfrac{vA(v)\big(A(u)^2-1\big)}{uA(u)} \end{bmatrix}$$

By the nature of M, we see that the diagonal elements of the transformation matrix are equal, $M_{00} = M_{11}$. Therefore, we require the equality of the elements. We equate:

$$A(u)A(v) + \frac{uA(u)\big(A(v)^2-1\big)}{vA(v)} = A(u)A(v) + \frac{vA(v)\big(A(u)^2-1\big)}{uA(u)} \tag{2.1.49}$$

Simplifying, we find that

$$\cancel{A(u)A(v)} + \frac{uA(u)\big(A(v)^2 - 1\big)}{vA(v)} = \cancel{A(u)A(v)} + \frac{vA(v)\big(A(u)^2 - 1\big)}{uA(u)}$$

$$\frac{uA(u)\big(A(v)^2 - 1\big)}{vA(v)} = \frac{vA(v)\big(A(u)^2 - 1\big)}{uA(u)} \tag{2.1.50}$$

$$\frac{\big(A(v)^2 - 1\big)}{v^2A^2(v)} = \frac{\big(A(u)^2 - 1\big)}{u^2A^2(u)} \tag{2.1.51}$$

The left hand side only depends on u, and the right hand side only depends on v. Since u and v are arbitrary, both sides must equal a universal constant.

Let that constant be k:

$$\frac{A^2(v) - 1}{v^2A^2(v)} = k \tag{2.1.52}$$

Rearranging, we have

$$A^2(v) - 1 = kv^2A^2(v)$$
$$A^2(v)(1 - kv^2) = 1$$

Therefore,

$$A = \frac{1}{\sqrt{1 - kv^2}} \tag{2.1.53}$$

where we choose the positive branch because the transformation must continuously approach the identity as $v \to 0$, thus reducing to the Galilean limit.

$$A(0) = 1 \tag{2.1.54}$$

Note the units of k. For A to be dimensionless, it has to have units which are $[L^{-2}][T^2]$, i.e, the inverse of the units of the square of velocity.

Now, notice that M_{10} reduces to

$$\frac{A^2 - 1}{vA} = kvA \tag{2.1.55}$$

Thus the transformation becomes

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = A \begin{bmatrix} 1 & -v \\ -kv & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \tag{2.1.56}$$

Equivalently,

$$x' = \frac{x - vt}{\sqrt{1 - kv^2}} \tag{2.1.57}$$

$$t' = \frac{t - kvx}{\sqrt{1 - kv^2}} \tag{2.1.58}$$

This is already the Lorentz transformation in generalized form.

From this, we can derive that

$$w = \frac{u + v}{1 + kuv} \tag{2.1.59}$$

This is the familiar relativistic velocity addition formula.

Writing k always in terms of $1/k^2$ is a bit cumbersome, especially when writing longer proofs. We hereby assign a new symbol for the speed. Let $\kappa = \frac{1}{\sqrt{k}}$.

Suppose we set $u = \kappa$. Then, we find

$$w = \frac{\kappa + v}{1 + kv\kappa} \tag{2.1.60}$$

Since $k\kappa = \dfrac{1}{\kappa}$, we then obtain

$$\begin{aligned} w &= \frac{\kappa + v}{1 + \frac{v}{\kappa}} \\ &= \frac{\kappa + v}{(\kappa + v)/\kappa} \\ &= \kappa \end{aligned}$$

thus:

$$u = \kappa \implies w = \kappa \tag{2.1.61}$$

The speed κ is therefore invariant under the velocity-composition law.

This is the crucial result from all that derivation. The principle of relativity does not tell us what the invariant speed is. It tells us that:

Theorem 2.1 : Universal invariant speed

For the Lorentzian branch of the kinematics, there must exist a universal invariant speed κ.

2.1.12 The Relevance of Light

Where did light enter the discussion? It has not entered the argument at all. We only began with the following fundamental principles

1. principle of relativity
2. homogeneity of space
3. homogeneity of time
4. isotropy of space
5. principle of inertia
6. closure of transformations under compositions

and these assumptions led us to find the transformation equations. Nowhere did Maxwell or electromagnetics come into the picture.

We did not need light at all. In this sense, the theory of relativity is derived and works on pure mathematical logic alone. All of physics bows down to this principle.

The empirical question that can then be raised is: Which value k describes our universe?

Experiment tells us that the invarient speed is the speed of light in a vacuum.

$$\kappa = c \tag{2.1.62}$$

so that

$$k = \frac{1}{c^2} \tag{2.1.63}$$

It did not need to be light. If we were more sensitive to gravitational waves, we would have called it thus. However, we are a species who evolved to sense light better, and it was this that lead us to stumble upon relativity. But there is in fact no need at all for it. A a blind society, insensitive to light, just with principles and logic, can establish relativity. Thus, electromagnetism does not need to be used to *derive* the Lorentz transformation. Instead, electromagnetism provides us an empirical way of identifying the invarient speed.

In this sense, light acts as a calibrator of the kinematics rather than as the logical foundation of the transformation.

2.1.13 The Galilean Limit

There is another possibility.

If we set $\kappa \to 0$, then $\kappa \to \infty$ and $A \to 1$, the transformation thus becomes

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} 1 & -v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

These are precisely the Galilean transformations. The corresponding velocity addition law is then

$$w = u + v$$

In this case, the invariant speed is effectively infinite. Thus, Galilean relativity can be viwed as the $\kappa \to \infty$ limit of the generalized transformation.

2.1.14 Lorentzian, Galilean, and Carrollian kinematics

The parameter k therefore distinguishes different possible kinematic structures.

For $k > 0$, there is a finite invariant speed

$$\kappa = \frac{1}{\sqrt{k}}$$

giving us Lorentzian relativity.

For $k = 0$, the invariant speed becomes infinite, giving Galilean relativity. The main reason that Galilean relativity took hold for so long is for quite some time in the enlightenment, people thought that the speed of light is infinite, and so that is the logical conclusion.

While inaccurate at relativistic scales, Galilean relativity, and by extension Newtonian physics, remains central to physics education because, for systems involving velocities $v \ll c$ and ordinary spatial and temporal scales, it provides an excellent approximation to relativistic physics. In this regime, the corrections predicted by special relativity are negligible, making the Galilean description both sufficiently accurate and considerably simpler as a description of reality.

There is also a limiting case in which the invariant speed tends to zero. As $k \to \infty$,

$$\kappa \to 0 \tag{2.1.64}$$

This is associated with the Carrolian limit of relativity. In this regime, the causal structure becomes increasingly restrictive: events at distinct spatial locations cannot influence one another within any finite time interval, and the notion of spatial propagation effectively disappears. The resulting Carrollian kinematics therefore describes a theory in which particles may evolve in time while remaining effectively frozen in space, providing the opposite extreme to the Galilean limit of instantaneous p“ropagation.

These three structures can therefore be understood schematically as:

`images/cones_types_of_relativity.pdf`

Figure 1: Lightcones in different types of relativity

The light cone structure which posits finite travel of information belongs to the finite Lorentzian case. The causal cone has a finite opening angle because there is a finite maximum invariant speed. In the Galilean limit, the cone opens out to incompase instantaneous propagation of information, no matter the distance of the interaction. In the Carollian limit, the cone collapses, corresponding to the extreme opposite situation where spatial propagation, and travel of information, is suppressed, and a body can only travel forwards in time, and no other.

2.1.15 Conclusion

This, and this is a point that I will not get tired telling you[9] proves the point that the derivation of the Lorentz transformation does not logically require the existence of light. The transformation

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \gamma \begin{bmatrix} 1 & -v \\ -\beta & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

can be obtained from the structure of inertial frames and the relativity principle alone, with the constant c initially appearing simply as an arbitrary invariant speed. Experiment then tells us that this invariant speed is

$$c = 299\,792\,458\,\mathrm{m/s} \tag{2.1.65}$$

and that electromagnetic wave propagation also happens to have an invariant speed, and that speed turned out to be c as well, for electromagnetic radiation travel precisely this speed in a vacuum.

The logical relationship can be summarized as

$$\text{Relativity} \implies \text{possible invariant speed } \kappa$$

while experiment estabishes

$$\kappa = c$$

Electromagnetism therefore provides an empirical realization of the invariant speed; it is not necessary as a premise in the kinematic derivation itself.

The deeper lesson is that the Lorentz transformations and relativity itself is fundamentally a statement about the symmetry of spacetime, not specifically about light.

[9] Actual verbatim quote from sir

Chapter 3

Aspects of Differential Geometry

Chapter 4

Einstein Field Equations

Chapter 5

Matter Models

Chapter 6

Weak-field Solutions

Chapter 7

Spherical Symmetry

Chapter 8

Basic Cosmology